Victor Chua

Singapore | +65 9190 5651 | chua1102@e.ntu.edu.sg | github.com/zeppyclown

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HP Inc., Singapore   Requisition UNI4651   13 August 2026

Dear Hiring Team,

I am applying for the College Intern position in Machine Learning and Artificial Intelligence. I am an NTU Computer Science undergraduate in the Turing AI Scholars Programme and can commit full time from January 2027 for five to six months, which matches the window in the posting.

Your first listed deliverable is an annotated dataset of print images. That is the part of this role I have already done end to end, on my own. No open labelled dataset existed for drum notation, so I built one: 19,948 labelled images from 264 source documents, 19 symbol classes and 10 duration classes, split by source document rather than at random so that no page could leak between train and test. I audited the labels across more than twenty rounds of thirty-sample random checks before training anything, and augmented with rotation, brightness and blur specifically so the model would survive skewed, faded and poorly scanned pages instead of only clean renders. Dealing with degraded print is not incidental to that project; it is most of it.

On the model, I trained a dual-head MobileNetV3-Small that classifies symbol type and duration at every position across a page in a single pass, reaching 90.9% duration accuracy and 97.4% per-cell accuracy on 1,924 held-out images from 26 sources the model had never seen. I documented the failure modes alongside the headline figures, exported to ONNX so inference runs locally at no per-image cost, and published the weights under MIT. I also built the review interface in TypeScript and Electron, because a recognition model that no human can inspect or correct does not get trusted.

I should be straight about the gap. I have not built a dedicated anomaly detection model. My recognition work is supervised classification, and detecting print artifacts, misprints and distortions is a different problem, closer to modelling what normal looks like and measuring departure from it. What I bring to it is the surrounding discipline: at my current internship I built an automated quality-assessment pipeline that scores recordings against a rubric, ranks them across a cohort, and regenerates a staff dashboard on every run, and I deliberately made it report which factor drove each low score rather than a bare number, because a reviewer needs the likely cause, not a verdict. That is the same instinct your root cause analysis deliverable asks for. At DBS I built the dashboard equivalent at scale, tracking memory and runtime across a pipeline of 3,000+ jobs a day broken down per version so a regression could be attributed to the release that introduced it.

I have also learned to distrust my own numbers. On a recent evaluation I found the headline accuracy had been measured on the same examples used to ground the judge, so it was not a blind test. I re-ran it on held-out data and reported both figures, even though the better-looking one was already circulating. On a project whose success criteria include a performance evaluation report, I would rather be the person who catches that early.

Thank you for your consideration. I would welcome the chance to discuss the project.

Yours sincerely,
Victor Chua